

Opinion

Population genetics, biofilm recalcitrance, and antibiotic resistance evolution

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Biofilms are communities of bacteria forming high-density sessile colonies. Such a lifestyle comes associated with costs and benefits: while the growth rate of biofilms is often lower than that of their free-living counterparts, this cost is readily repaid once the colony is subjected to antibiotics. Biofilms can grow in antibiotic concentrations a thousand times higher than planktonic bacteria. While numerous mechanisms have been proposed to explain biofilm recalcitrance towards antibiotics, little is yet known about their effect on the evolution of resistance. We synthesize the current understanding of biofilm recalcitrance from a pharmacodynamic and a population genetics perspective. Using the pharmacodynamic framework, we discuss the effects of various mechanisms and show that biofilms can either promote or impede resistance evolution.

Biofilms and the evolution of antibiotic resistance

Biofilms, the predominant form of bacterial life, are spatially structured, heterogeneous aggregates of bacterial cells that can be visible to the naked eye [1,2]. Their particular physiology enables them to survive in various unfavorable conditions. Despite its obvious importance, however, little is known about the effect of the biofilm lifestyle on the evolution of antibiotic resistance [3,4]. Here, we approach this issue from a population genetics perspective. First, we investigate the effects of individual mechanisms of biofilm **recalcitrance** (see [Glossary](#)) on population growth rate and formalize them using **pharmacodynamic functions**. Then we illustrate and discuss their consequences for population dynamics and the rate of resistance evolution.

Despite the prevalence of biofilms, experimental studies are usually carried out in planktonic cultures. The wealth of data on planktonic lifestyles dwarfs the available knowledge on biofilms [1]. This recently started to change, as biofilms were recognized to be a significant contributor to many bacterial diseases: almost 80% of all bacterial infections in humans are a direct consequence of biofilms [5], including cystic fibrosis, urinary tract infections, and even cancer development [6,7]. Biofilms can cause chronic wound infection, leading to limb amputations in diabetic patients [8]. In addition, multiple diseases are linked to exogenous devices and implants, such as catheters, heart valves, and prosthetic joints [9].

Biofilms contribute to the pathogenesis of diseases through many different mechanisms, creating physical barriers (e.g., in heart valves) or serving as reservoirs for pathogenic bacteria that are shed into the body, causing repeated acute infections [2,10,11]. Even the mere presence of biofilms may stimulate long-term inflammatory responses, resulting in tissue damage [2,11]. Biofilms are often too big to be ingested by neutrophils [12] and can survive and grow at high antibiotic concentrations that are toxic to the treated organisms. Thus, biofilm infections are often pathogenic and usually chronic and difficult to clear by the immune system or antibiotics [13,14].

Highlights

Most of the mechanisms of biofilm recalcitrance proposed in the current literature can be modeled by adjusting pharmacodynamic functions.

In biofilms, protection mediated by reduced penetration decreases the selection for resistant mutants, resulting in resistance evolution that is remarkably slower than in planktonic cultures.

The same protection enables antibiotic resistance to evolve in biofilms in a very high antibiotic concentration regime, which leads to the fast extinction of planktonic populations.

Physiological alterations of biofilm cells (i.e., dormant or persister cells) slow down the evolution of resistance. However, these mechanisms ensure population survival if antimicrobials are degraded at a sufficiently high rate.

The combination of physiological alterations and reduced penetration is predicted to lead to the most prolonged survival of bacterial populations and recalcitrance against the highest antibiotic concentrations.

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The formation of biofilms starts with a few bacterial cells attaching to a substrate. Then the bacteria begin to exude an extracellular polymeric substance (EPS), which together with host components forms the extracellular matrix. It contains mainly polysaccharides, proteins, nucleic acids, and lipids, but structure and composition vary greatly between biofilms, depending on the bacterial strains, nutrient composition, location, and physical conditions such as temperature and shear forces [10,15,16]. The biofilm then grows and matures, becoming spatially structured and layered, and forms water channels [17]. The extracellular matrix immobilizes bacterial cells in a dense network, providing mechanical stability. By preventing extracellular enzymes from dispersing, it enables the extracellular digestion of various substances and increases concentrations of enzymatic defenses against antibiotics [18]. By acting as a physical barrier or by chemical reaction, the extracellular matrix protects bacteria not only from the immune system but also from antibiotics, biocides, UV light, or charged toxic compounds [15].

This multilayer, large spatial structure causes varying environmental conditions that the cells inside the biofilm encounter. They may suffer from hypoxia and reduced concentrations of nutrients. As a result, phenotypic heterogeneity is high within biofilms [10]. The maximum cell density in a biofilm is regulated by quorum sensing, forming a community that can harbor multiple species of bacteria. These can form stable micro-consortia with frequent horizontal gene transfer and intense cell-to-cell communication [15]. Cells disperse or actively detach from mature biofilms and can live in a planktonic state or form new biofilms [19].

One of the defining and clinically most relevant features of biofilms is their ability to survive and grow in high antibiotic concentrations, denoted recalcitrance. Mature biofilms are known to proliferate in antibiotic concentrations up to three orders of magnitude higher than those killing planktonic cells [20]. In recent years, numerous explanations for the observed recalcitrance of biofilms have been proposed and thoroughly reviewed [11,12,14,21–30], together with strategies to combat it [29]. Rather than reviewing them again, we focus on the effects of these mechanisms on population growth rates and show how they can be quantitatively captured in established pharmacodynamic models. Then we discuss their consequences for the genetic composition of the bacterial populations, focusing on the evolution of **antibiotic resistance**. Note that we only consider single-species biofilms, as the complexity of the species interactions is outside the scope of this article; for recent reviews of recalcitrance in multi-species biofilms, see those of Orazi *et al.* [31] and Sadiq *et al.* [32].

Effects of recalcitrance mechanisms on pharmacodynamics

The recalcitrance of biofilms increases with increasing maturity and thickness [33]. However, cells that disperse from the biofilm are usually susceptible to antibiotics, often to the same degree as the original planktonic population [23]. Therefore, mechanisms that are a direct consequence of the biofilm lifestyle must play an important role in biofilm recalcitrance. Such a phenotypic, nonheritable ability to survive antibiotic treatment is sometimes termed tolerance, though the meaning of tolerance in this context is slightly different from that in planktonic cells. The part of the recalcitrance that is retained by the dispersed cells is primarily genetic and thus heritable [34]. Even though it is usually smaller than the phenotypic component, it has important implications for the infection treatment success and may lead to heritable antibiotic resistance.

Genetic aspects of biofilm recalcitrance

Like free-living bacteria, cells in biofilms mutate and can acquire mutations beneficial in the environment containing antimicrobials. Mutations conferring antibiotic resistance are generally considered to be costly to the carrier in the absence of the antibiotic (note, however, that this trade-off is currently investigated, yet not entirely determined [35,36]). For instance, a mutation

Glossary

Antibiotic resistance: in this article, antibiotic resistance is any genetic alteration (by mutation or horizontal gene transfer) that increases minimum inhibitory concentration (MIC) of the carrier above the value of the wild type. Note that in clinical microbiology, resistance usually is defined as bacterial growth in antibiotic concentrations above a given threshold, a breakpoint determined by The European Committee on Antimicrobial Susceptibility Testing (EUCAST). These breakpoints vary across species and are reassessed regularly and sometimes redefined.

Mutant selection window (MSW): the concentration range where the growth rate of the mutant is positive and higher than the growth rate of the wild type.

Persister cells: bacterial cells that have a vanishingly low division rate but are able to survive high concentrations of antibiotics.

Pharmacodynamic function: describes how the bacterial population growth rate ψ (which can be negative) depends on the antibiotic concentration c . Its reverse sigmoidal shape (see Figure 1 in Box 1) is defined by the maximal bacterial growth rate in the absence of antibiotic ($\psi_{\max}$); minimum inhibitory concentration (MIC) where the growth rate crosses the horizontal axis and becomes negative with increasing antibiotic concentration; the Hill coefficient (k) determining the steepness of the curve; and an additional parameter capturing the rate of the lowest net growth rate ($\psi_{\min}$), a rate that can be attained at very high antibiotic concentrations [78]:

$$\psi = \psi_{\max} - \frac{(\psi_{\max} - \psi_{\min}) \left(\frac{c}{\text{MIC}} \right)^k}{\left(\frac{c}{\text{MIC}} \right)^k + 1} \quad [1]$$

Recalcitrance: the ability of biofilms to survive and grow in high antibiotic concentrations.

in a drug-binding site may protect a protein from interaction with a drug [37], but it also impairs its function, reducing the growth rate of the mutant. Therefore, unless other compensatory mutations reducing the cost occur [38], antibiotic resistance mutations are typically purged from the population in the absence of an antibiotic or present only at low frequencies. However, the cost is repaid in the presence of an antibiotic: while the wild type mutant suffers the drug's effect, the mutant growth rate may exceed that of the wild type, possibly leading to the spread of the resistance mutation.

The effect of mutations on bacterial growth can be modeled by adjusting the pharmacodynamic function: fitness costs are represented by the reduction of the maximum growth rate $\psi_{\max}$, while an increase in the minimum inhibitory concentration (MIC) needed to stop the bacterial growth captures the benefit. Therefore, the shape of the growth-defining function is changed: the curve typically starts at a lower growth rate and is shifted to the right [39], compared with the sensitive strain (Figure 1, Key figure).

The concentration range where the growth rate of the mutant is positive and higher than the growth rate of the wild type is called the **mutant selection window (MSW)** [40–42]. The intersection of the curves defines the lower boundary of the MSW: it is the lowest concentration at which the mutant's growth rate is above the growth rate of the wild type. The intersection of

Key figure

Biofilm lifestyle and pharmacodynamics

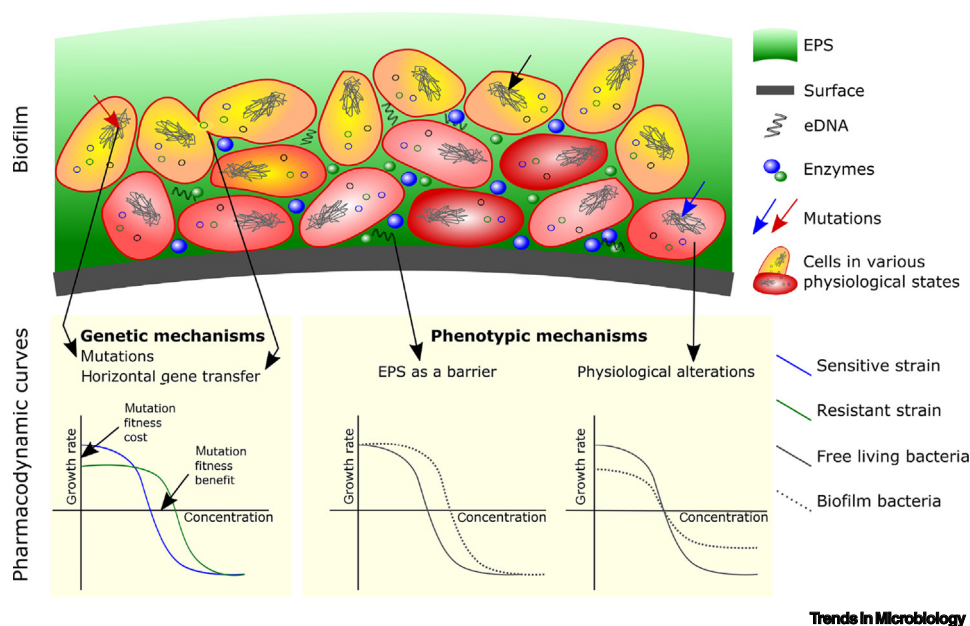


Figure 1. Many aspects of the biofilm lifestyle influence its recalcitrance to antibiotics. Genetic mechanisms (mutations, horizontal gene transfer) give rise to resistant mutants. The extracellular matrix and its components may act as a barrier, preventing antibiotics in the medium from reaching their targets, effectively increasing MIC. Physiological modification of a large fraction of the population reduces the population's sensitivity to the drug treatment, thus reducing the rate of death. All of these mechanisms alter the pharmacodynamic properties of the populations, shifting and flattening pharmacodynamic curves. Abbreviations: eDNA, extracellular DNA; EPS, extracellular polymeric substance; MIC, minimum inhibitory concentration.

the pharmacodynamic function of the least drug-susceptible mutant with the x-axis (mutant's MIC) is called mutant prevention concentration and defines the upper boundary of the MSW (see Figure 1A in Box 1) [40,41]. Thus, MSW denotes a concentration range where the mutants are selected for, potentially leading to the spread of resistance [41]. As discussed later, the MSW can be shifted and distorted by the effects of nongenetic recalcitrance mechanisms present in biofilms.

All of the aforementioned genetic mechanisms apply to both free-living and biofilm cells. However, several types of mutations may only decrease the susceptibility of biofilm-inhabiting cells (e.g., mutations influencing the production of extracellular matrix).

Nongenetic mechanisms of biofilm recalcitrance

As bacteria dispersed from biofilms are usually susceptible to antibiotics, there must be additional, nongenetic mechanisms inherent to the biofilm lifestyle that increase the recalcitrance of biofilm-inhabiting bacteria towards antibiotics. Numerous explanations for such nongenetic recalcitrance have been proposed [11,12,21–23,25–30]. Most of them can be broadly characterized by their underlying features into those that act as a barrier and reduce the antimicrobial concentration and those that rely on physiological alterations.

Box 1. Shifts and distortions of mutant selection windows (MSWs)

The drug response of a bacterial strain is determined by its genotype and the mutations it carries, as well as by its lifestyle. The MSW (shaded areas in Figure 1) is defined by the pharmacodynamic functions of the wild type (blue) and the mutant (green) bacterial strains [36,41]. However, both of these functions can be altered in biofilms, as shown in Figure 1 in the main text: they can be either shifted to the right, for instance, when the number of drug molecules reaching the target is reduced, effectively increasing MIC (e.g., by EPS acting as a barrier); or flattened, if part of the population (e.g., persisters) does not reproduce and does not react to the drug.

If the pharmacodynamic curves are shifted, the MSW in a biofilm is shifted to higher antibiotic concentrations and may be wider (or narrower) than the original MSW in planktonic cells, which depends on the particular mutation, as well as recalcitrance mechanism involved and how these effects combine. Moreover, both windows may or may not overlap, which, in turn, depends on the relative benefits of resistant mutations versus biofilm lifestyle.

However, if the pharmacodynamic curves are flattened (e.g., due to the presence of persisters, dormant cells, or cells with reduced metabolism or growth rate), MIC is unchanged, leaving the upper boundary of the MSW unchanged. Whether the lower boundary of MSW changes depends, again, on the combined effect of mutations and altered physiology. If we assume that physiological changes are independent of the genotype (thus equally affecting mutants and sensitive wild types), pharmacodynamic curves are flattened equally and the MSW stays the same. This is, however, unlikely: the curves can be influenced in different ways, distorting the MSWs.

Nevertheless, in natural biofilms, both types of mechanisms likely play a role, thus shifting and flattening the pharmacodynamic curves simultaneously, resulting in significant alterations of the MSW.

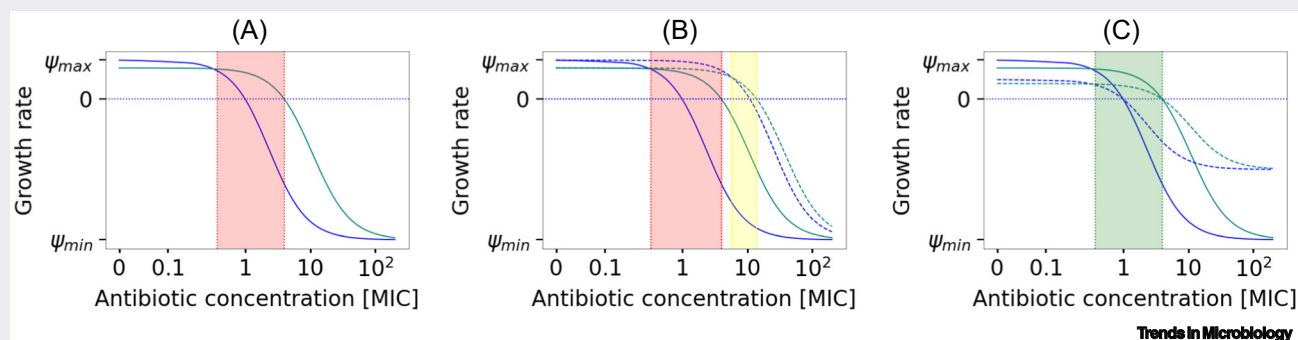


Figure 1. Effect of biofilm lifestyle on mutant selection windows (MSWs). Blue lines represent wild type, green lines mutants, adopting either planktonic (unbroken lines) or biofilm (broken lines) lifestyle. Note that the scale of the x-axis is logarithmic. (A) The mutant selection window in plankton (pink). (B) Pharmacodynamic functions are shifted due to the shielding of the drug targets, resulting in the shift of the MSW (yellow) compared with MSW in plankton (pink). In this example, both windows are approximately equally wide. (C) Growth of a population containing a high fraction of persisters. Abbreviation: MIC, minimum inhibitory concentration.

Physical and chemical barriers

The extracellular matrix provides a physical barrier for larger substances, preventing their penetration deep into the biofilm. Therefore, one of the first explanations for the biofilm recalcitrance was based on restricted antibiotic penetration. However, most antibiotics are small enough to penetrate deep into the biofilm. Some antimicrobials may be depleted due to the interaction with extracellular matrix components of the biofilm, such as extracellular DNA [11,23,43], or degraded by antibiotic-modifying enzymes (e.g., beta-lactamases) in the extracellular matrix [43–45]. Furthermore, [46] suggested that depletion of the antimicrobials by the fluid surrounding the biofilm or the substrate it grows on, either by reaction or sorption, may contribute to the observed recalcitrance.

All these mechanisms reduce the number of antibiotic molecules that reach the bacterial cell, meaning that bacteria in the biofilm are often exposed to lower concentrations of antimicrobials [47] than what is measured outside of the biofilm. This can be perceived as an increase in the MIC of the biofilm-inhabiting bacteria and modeled by shifting the pharmacodynamic function to the right (Figure 1).

This shift influences both wild types and mutants, though not necessarily by the same amount. This also implies that the concentrations at which resistance-conferring mutations are beneficial (i.e., the concentration at which the mutant's growth rate exceeds that of the wild type) will be shifted. Thus, the entire MSW moves to higher concentrations. As the wild type and the mutants may be affected differently, the width of the MSW can change (see Figure 1B in Box 1).

While physical and chemical properties of biofilms can act as barriers for some classes of antibiotics (e.g., aminoglycosides), most antibiotic molecules are small enough to penetrate the biofilm [48,49] and do not react chemically with extracellular matrix components [12,50–52]. Anderl *et al.* [51] observed, for example, that ampicillin, which is degraded by beta-lactamase (present in the EPS), did not significantly penetrate wild-type *Klebsiella pneumoniae* biofilms. By contrast, ciprofloxacin penetrated the biofilms quickly, reaching the MIC throughout the biofilm within 20 minutes. However, when the lactamase-deficient mutant was treated with ampicillin, the drug penetrated but did not kill the biofilm. These results indicate that reduced penetration is biofilm- and antibiotic-specific and that biofilm recalcitrance cannot be fully accounted for by antibiotic inactivation or low penetration alone [51].

Physiological alteration

Another set of explanations of biofilm recalcitrance is based on changes in metabolism, growth rate, or the more general physiological state of the biofilm bacteria. Many antibiotics target mechanisms or processes that occur in growing bacteria (including replication, protein production, cell wall synthesis); nondividing or slow-dividing bacteria offer fewer of these targets and are thus less sensitive to antibiotics [11,53]. Therefore, the efficacy of many antibiotics is reduced on nongrowing or slowly growing bacteria [5].

Due to phenotypic differences in growth and division rates, the relationship between division rate and sensitivity is even relevant within clonal bacterial populations. In a recent study, Lee *et al.* [54] measured β -lactam-mediated lysis rate in *Escherichia coli* in response to the growth rate and found a positive linear relationship between growth and lysis rates for diverse antibiotics. Subpopulations with extremely reduced growth rates have been identified [55,56] and, because they are refractory to antibiotics, are called persisters [14,57]. Because the recalcitrance of persisters is based on phenotypic, nonheritable growth characteristics, **persister cells** become fully sensitive again after they switch back to active growth.

Heterogeneity of environmental conditions within the biofilm causes cells to adopt different phenotypes, some of which are less sensitive to antibiotics than others. Cells deep in the biofilm are deprived of oxygen and nutrients, leading to reduced metabolic activity and growth [11,12,14,44]. Several studies suggest that reduced growth rates resulting from nutrient limitation are partly responsible for biofilm recalcitrance [50,52,58], which was supported by a modeling study [59]. Vrany *et al.* [50] suggest that the bacteria within biofilms, even if not nutrient-limited, may grow at a reduced rate compared with planktonic cells due to a higher cell density. Spoering and Lewis [52] argue that in *Pseudomonas aeruginosa*, persisters, stationary state cells, and biofilms are effectively the same. Analyzing the death rates of planktonic cells in an exponential phase, stationary state, and biofilms, the authors show that killing was the same for stationary and biofilm cells but much higher for cells in the exponential phase. The authors' reason that the fraction of persisters in the population is regulated by cell density and increases in the stationary phase and biofilms compared with cells in the exponential phase. In any case, biofilms contain cells in various physiological states: growing, tolerant, and stress-adapted with reduced growth, even completely dormant [12,46,60,61]. This heterogeneity plays an important role in biofilm recalcitrance.

The persistent cells are able to survive high concentrations of antibiotics but do not divide; their growth rate is 0. As the population is a mixture of cells, most of whom retain the wild type pharmacodynamic properties, the resulting pharmacodynamic function of the whole population is an average of the two types. The MIC of the population is unchanged compared with the sensitive population [57]. However, the maximum growth rate in the absence of antibiotics is significantly reduced and the minimum (negative) net growth rate in the presence of high antibiotic doses is increased. A population containing a large fraction of persisters can be thus modeled by flattening the pharmacodynamic function, as shown in Figure 1.

Whether the population consists of sensitive wild types or resistant mutants, the pharmacodynamic curves are changed, possibly altering the MSW. The MIC is unchanged, leaving the upper boundary of the MSW unchanged. If we assume that physiological changes are independent of the genotype, pharmacodynamic curves are flattened equally; their intersection (the lower boundary of MSW) does not shift and the MSW stays the same (see Figure 1B in Box 1). This, however, only applies when sensitive and resistant mutants are affected in the same way by physiological changes or when the fraction of persisters is the same in the sensitive and resistant population.

In either case, both types of nongenetic recalcitrance mechanisms alter pharmacodynamic curves of sensitive and resistant bacteria, thus affecting the population dynamics with profound consequences for antibiotic resistance evolution, which we discuss later.

Consequences of biofilm lifestyle for antibiotic resistance evolution

Biofilm growth is typically assumed to be slower than the growth of the planktonic population. This suggests that opportunities for mutations are less abundant, potentially slowing the resistance evolution. However, some studies suggest that biofilm lifestyle and antibiotic presence can induce a stress response leading to an increase in mutagenesis [47], compensating for the slow growth. Furthermore, horizontal gene transfer can occur through the transfer of plasmids between cells in a biofilm via conjugation at a higher frequency than in planktonic cells due to the close contact between the cells [23]. In addition, some bacteria can pick up extracellular DNA from the biofilm matrix. All these mechanisms suggest an increased probability of antibiotic resistance evolution. As the resistance mechanisms we are considering have a genetic basis and are heritable, they are retained by the bacterial cells upon dispersal, contributing to the spread of antibiotic resistance, with profound consequences for health care and agriculture [34].

Knowing how the biofilm lifestyle affects resistance evolution is not intuitive. As discussed in the previous sections, multiple nongenetic mechanisms are involved in biofilm recalcitrance that alter the expression of the genotype and interfere with evolutionary processes. Biofilms can act as shields, protecting the bacteria from antibiotics, reducing selection pressure, and hence slowing the evolution of resistance. However, the same protection opens the opportunity for resistance evolution at antibiotic concentrations that would be fatal for planktonic bacteria. Similar conflicting evolutionary pressures also exist in planktonic settings. For instance, Levin-Reisman *et al.* [62] have shown that tolerance increases the probability of antibiotic resistance evolution by helping the population to survive longer, offering more opportunities for a resistance mutation to emerge. Other phenotypic effects, such as priming and persistence, can also influence the probability of resistance evolution [63].

Studies of antibiotic resistance evolution typically focus on planktonic bacteria (but see [64]), even though this is not usually explicitly mentioned, especially not in theoretical studies. Nevertheless, understanding interactions between phenotypic and genotypic factors influencing biofilm recalcitrance is crucial for maximizing the probability of successful treatment and minimizing the risk of antibiotic resistance evolution. In the following sections, we relate the changes of the MSWs to the consequences of such nongenetic mechanisms on the genetic composition of bacterial populations and the evolution of antimicrobial resistance.

Effect of physical and chemical barriers

MSW describes a range of concentrations, where the mutant has a fitness advantage compared with the resident, wild-type population. As discussed earlier, if the biofilm lifestyle hinders the access of antibiotics to their targets, pharmacodynamic curves for wild type and mutants are shifted to the right, resulting in the shift and distortion of the MSW. This shift has profound consequences for the evolution of antibiotic resistance.

On the one hand, the added protection of the biofilm lifestyle serves as a shield, decreasing the selective advantage of the mutants. Thus, the wild-type genotype is beneficial in a broader range of antibiotic concentrations, slowing down the evolution of resistance (Figure 2, ii and iii). The shift of MSW in biofilms gives rise to a range of concentrations, where a mutation is already beneficial in the plankton but still detrimental in the biofilm lifestyle (see Figure 1B in Box 1, pink area). In such a range of concentrations, the resistant mutant will quickly replace the sensitive wild type in plankton but not in biofilms, where resistance mutations only start to be beneficial at higher concentrations.

As already demonstrated experimentally in biofilms produced by *Acinetobacter baumannii* in the presence of ciprofloxacin [4], the difference between the planktonic and biofilm lifestyles becomes clearly visible in evolutionary experiments where populations are gradually challenged by increasing antibiotic concentrations and multiple mutations are involved in resistance evolution. Increasing selective pressure for antibiotic resistance can dramatically speed up its evolution, as discussed in Box 2. Furthermore, reduced selection pressure experienced by biofilm-inhabiting cells will reduce clonal interference, enabling multiple mutants to coexist, increasing the diversity (Box 2).

On the other hand, by shielding the bacterial population from the killing effects of antibiotics, the biofilm lifestyle enables the evolution of resistance at high concentration, where planktonic bacteria, despite carrying a beneficial mutation, would be killed (Figure 2, iv). Again, the shift of MSW in biofilms (see Figure 1B in Box 1, yellow area) gives rise to a range of concentrations, where a mutation confers a sufficient advantage for the bacteria to survive in biofilms, but not sufficient to survive when living freely. Thus, high doses of antibiotics may clear the acute infection, but

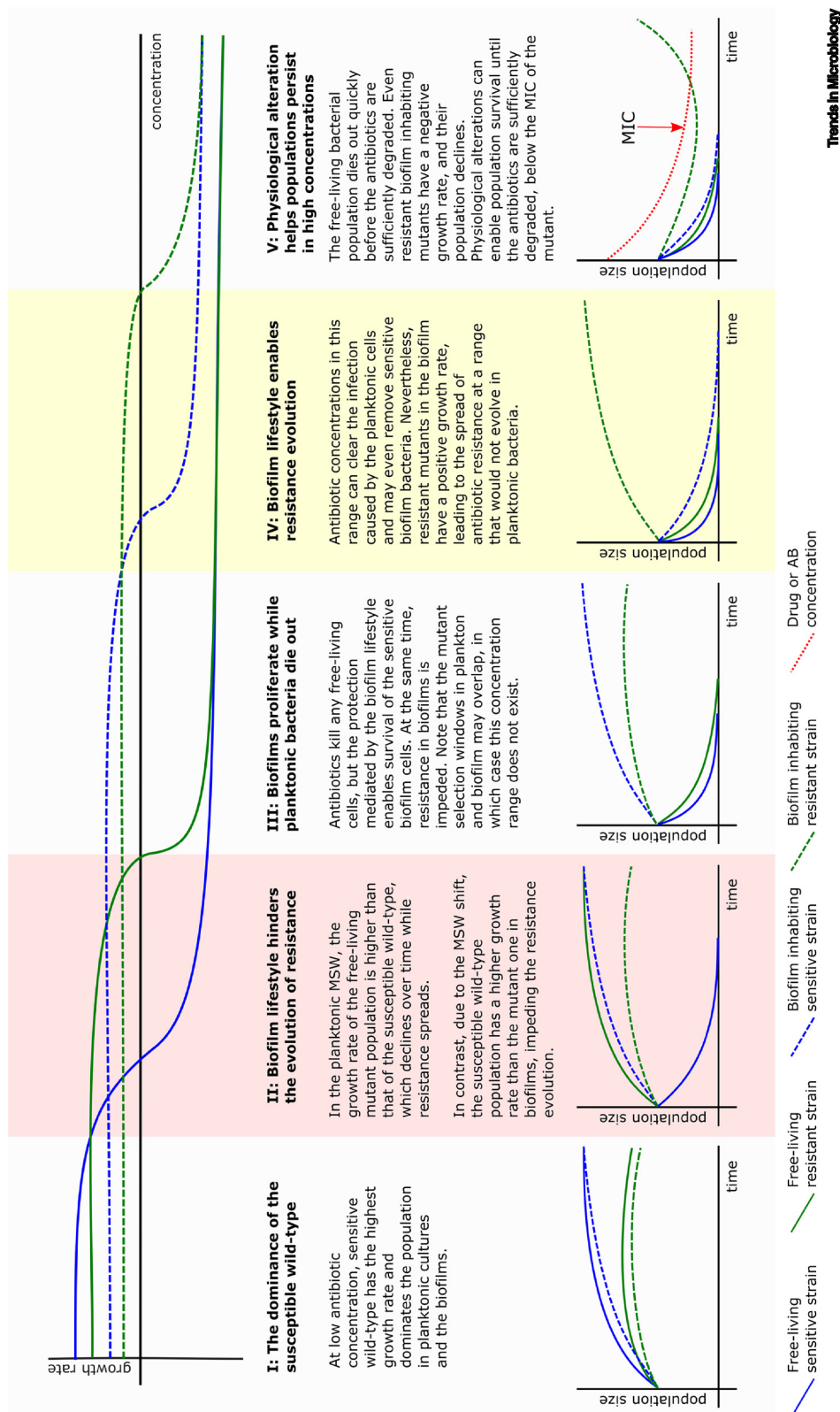


Figure 2. Consequences of recalcitrance mechanisms for the evolution of antibiotic resistance. The effects of various recalcitrance mechanisms on the evolution of resistance depend on their effects on pharmacodynamic curves and differ at various concentrations. Pharmacodynamic curves of sensitive and resistant bacteria adopting free or biofilm lifestyle determine five key concentration ranges (left) that differ in their consequences for population dynamics and resistance evolution. Pink area: mutant selection window (MSW) in plankton; yellow area, MSW in biofilms. All populations start at the same size. Competition occurs only between the sensitive and resistant strains, not between the biofilm and plankton. Abbreviations: MIC, minimum inhibitory concentration.

Box 2. On multiple mutations and diversity

Theoretical studies of antibiotic resistance evolution traditionally assume that a single mutation mediates sufficient resistance [42,68–70]. However, mutational effects of resistance-conferring mutations can be of various sizes and multiple mutations can be combined [71]. Both costs and benefits of individual mutations have been shown to combine in a multiplicative manner [72,73]. However, it has also been shown that the probability of antibiotic resistance evolution by a single large mutation is larger than the evolution of resistance mediated by multiple mutations of smaller effects, which gets substantially limited if more than two mutations are required [36]. This provides a compelling reason for the combinations of various drugs to minimize the probability of treatment failure (but see [74]).

The difference in the rate of resistance evolution between the planktonic and biofilm lifestyles becomes the most obvious when multiple mutations are necessary for adaptation to higher doses. This occurs, for instance, in evolutionary experiments, when populations are gradually challenged by increasing antibiotic concentrations. With escalating concentrations, the selective pressure for resistance mutations increases, leading to quick fixations of beneficial alleles: sweeps. As multiple mutations are necessary for survival in the higher concentration, the average number of mutations per bacteria will rise steeply in planktonic cells. However, the cells protected by the biofilm lifestyle will not experience such selective pressure; at first, no mutations are necessary for survival, or only a few are required. Thus the average number of mutations per cell will increase slowly at first, or not at all. Only at higher concentrations, where biofilm-mediated protection is no longer sufficient, does the prevalence of mutations rise among biofilm cells (Figure I).

The number of mutations conferring resistance also affects the genetic diversity. While the diversity of resistance mutations is likely higher in plankton at low antibiotic concentrations (in the planktonic MSW), it decreases as the competition among mutants becomes more severe and only bacteria carrying multiple mutations are able to survive, replacing the less resistant variants (selective sweeps occur). By contrast, various single mutations are still able to confer sufficient resistance in biofilms at such concentration ranges and multiple mutants may coexist. The higher genetic diversity in biofilms, compared with planktonic cultures (especially with respect to resistance mutations), has been observed experimentally multiple times [7,75]. The rigid spatial structure of biofilms further reduces the competition between various mutants, increasing the genetic diversity of biofilms even further.

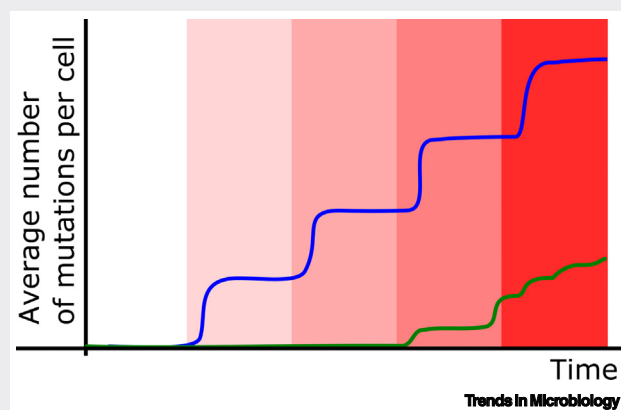


Figure I. Average number of mutations per cell increases faster in plankton than in biofilm. Average number of mutations per bacterial cell in plankton (blue) and in biofilm (green), in an increasing antibiotic concentration, indicated by the shades of red.

can also stimulate the evolution of resistance in a surviving biofilm, leading to another, now untreatable, infection with resistant bacteria. The antibiotic concentration necessary to prevent resistance evolution in biofilm is much higher and possibly toxic for the patient; the most notable example being the cystic fibrosis model of biofilm-related infection [65–67].

Effect of physiological alteration

As discussed earlier, an increased fraction of persisters or cells that are dormant, or growing slowly, does not increase the MIC of such a population [57]. Therefore, this mechanism itself would not explain the growth of biofilms at concentrations above the MIC. Nevertheless, it

can explain its prolonged survival. When combined with the degradation of antibiotics over time, reduction of growth rates and subsequent reduction in killing rates may help bacteria survive until the antibiotic is sufficiently degraded. Therefore, not only the amount of antibiotics but also their pharmacokinetic properties (e.g., half-life) determine the survival and growth of biofilms. Regrettably, this factor is often omitted from biofilm recalcitrance studies and mathematical models of resistance evolution [76].

The physiological alterations may play an especially important role at high antibiotic concentrations when combined with other recalcitrance mechanisms. At high concentration, all bacteria, whether wild type or mutant, living either as plankton or as biofilm, have a negative growth rate: the rate of death due to antibiotics exceeds the replication rate. The growth of the free-living bacterial population is affected by the drug's presence the most and, in most cases, the bacteria die out quickly before the antibiotics are sufficiently degraded. By comparison, a slowly growing biofilm population is less affected by antibiotics. Reduced growth rate and, subsequently, death rate may enable survival until the antibiotic is sufficiently degraded, below the threshold inhabiting the biofilm growth. Even though its size declines for a while, biofilm survives long enough until the drug is degraded, upon which it resumes its slow growth. As mutants are able to restart their growth sooner, at higher antibiotic concentration, this mechanism may facilitate the resistance evolution (Figure 2, v). Thus, accounting for pharmacokinetic properties of antibiotics is essential, especially at this concentration range, and should be taken into account in treatment design.

While both types of phenotypic recalcitrance mechanisms, either the effect of the barrier or altered physiology, are able to facilitate the evolution and spread of the antimicrobial resistance, they most likely act in combination, leading to the ability of biofilms to survive and thrive in extremely high antibiotic concentrations. However, the relative contributions of individual mechanisms likely differ among strains, drugs, and biofilm growth conditions and thus need to be assessed individually in each particular scenario. Nevertheless, our illustrations show that while both types of mechanisms extend the survival of biofilms in different ways, their combination brings the most substantial advantage.

Concluding remarks

Here, we discussed the evolution of antibiotic resistance in biofilms from the perspective of population genetics and pharmacodynamic modeling. While one major group of mechanisms is based on biofilm acting as a barrier, reducing the effectiveness of antibiotics, the other relies on physiological changes that help the cells survive until the drug is sufficiently degraded. We discussed how these types of mechanisms influenced pharmacodynamics and formalized this modification using the pharmacodynamic functions, which translated into distortions and shifts of the MSWs. Then we discussed and illustrated the population dynamic and genetic consequences of these recalcitrance mechanisms in isolation and combination. Specifically, we elucidated how they contribute to biofilm survival and resistance evolution.

The precise quantitative contributions of each of these factors, however, depend on the specific antibiotics, bacterial strain, as well as environmental and growth conditions [23,46] and need to be determined in future studies. Furthermore, numerous other aspects of biofilm lifestyle are necessary to create a complete picture of recalcitrance (see Outstanding questions): for instance, spatial structure creating diverse niches and allowing various phenotypic specializations, or species interaction, where a presence of one species may change the sensitivity profile of the other species, influencing the success of antimicrobial

Outstanding questions

What are the relative contributions of different mechanisms to biofilm recalcitrance in different bacterial species?

What is the importance of such different mechanisms and what are their implications for the evolution of antibiotic resistance?

What is the importance of the spatial aspects in biofilm recalcitrance and what are its implications for the evolution of antibiotic resistance?

What is the effect of diversity in multispecies communities?

Can we use population genetics models to predict the effects of various treatment strategies on combating antibiotic resistance evolution and biofilm eradication?

Ultimately, can we identify optimal treatments of recalcitrant biofilms that, while clearing the biofilm, do not select for additional resistance?

treatment [77]. Additionally, there is a higher interspecies horizontal gene transfer; further, extracellular enzymes or quorum sensing molecules produced by specific species tend to have a cumulative effect on the whole poly-microbial biofilm, complicating infections and their modeling [31].

The recalcitrance of biofilms to antibiotics is a multifactorial phenomenon. Each of the mechanisms we considered can, in isolation, only partially explain the reduced sensitivity of biofilm-inhabiting bacterial cells. Acting together, they can increase the MIC by three orders of magnitude, ensuring the survival of biofilms under the most aggressive treatments. Understanding their interactions and consequences for antibiotic resistance evolution is a major challenge, with profound consequences for human health.

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Declaration of interests

No interests are declared.

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